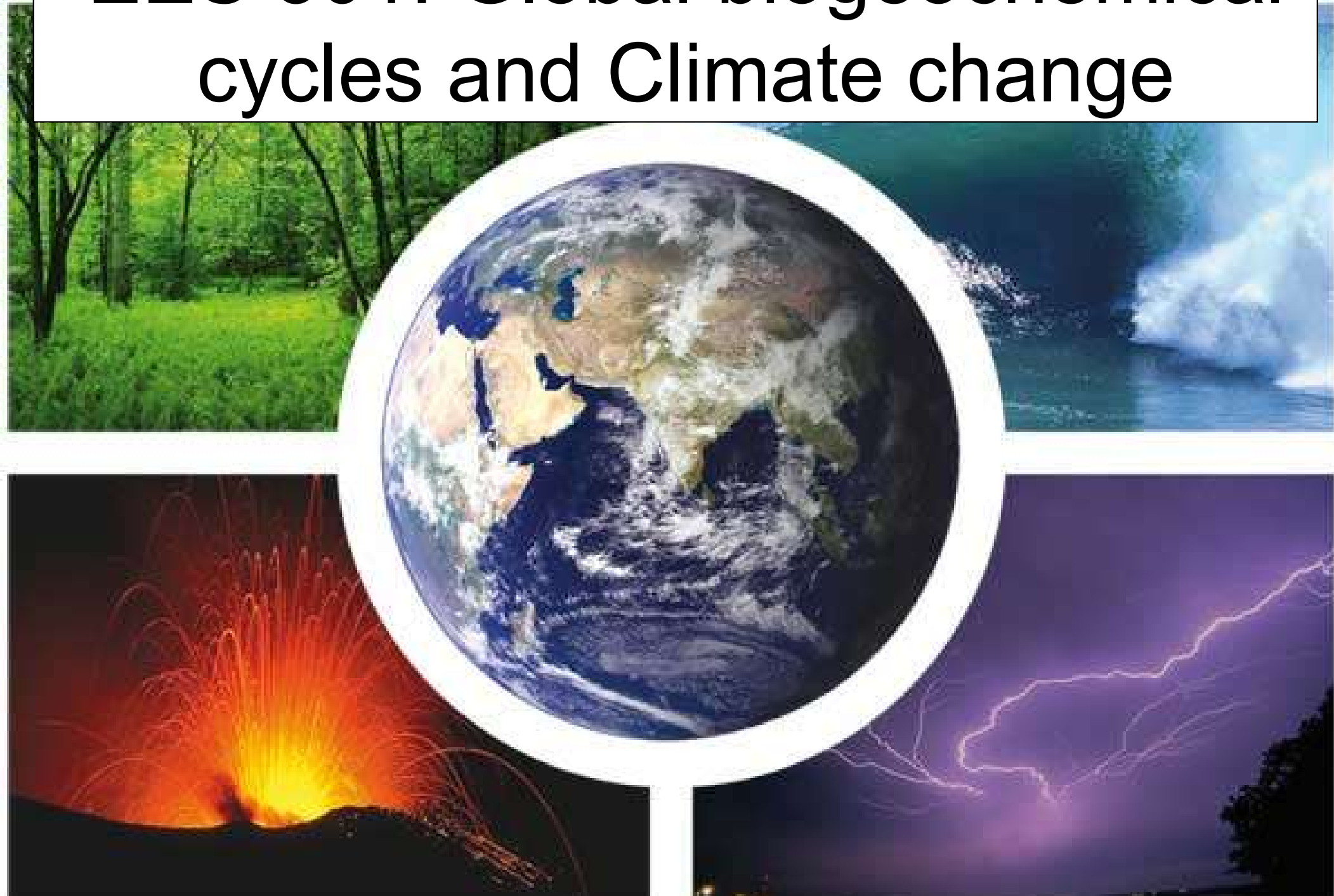


EES 301: Global biogeochemical cycles and Climate change



The Ocean

The ocean is the largest reservoir of most substances and elements within the biogeochemical cycles. Hence the study of transport processes within the ocean into the ocean and from the ocean to other reservoirs are central to the study of most biogeochemical cycles.

The topography of the ocean floor is created by the plate tectonic system. Oceans have a continental margin with a shelf. The shelf break tends to occur roughly at 200 m below the current sea level on average 70 km from the shoreline. During glacial maxima of the Pleistocene the sea level is 121 ± 5 m lower and much of the shelf becomes a coastal plain. The continental margin will lead to a shelf-slope-rise basin along most oceanic continental plate margins and to a shelf-slope-trench margin on plate boundaries where an oceanic plate subducts under a continental plate

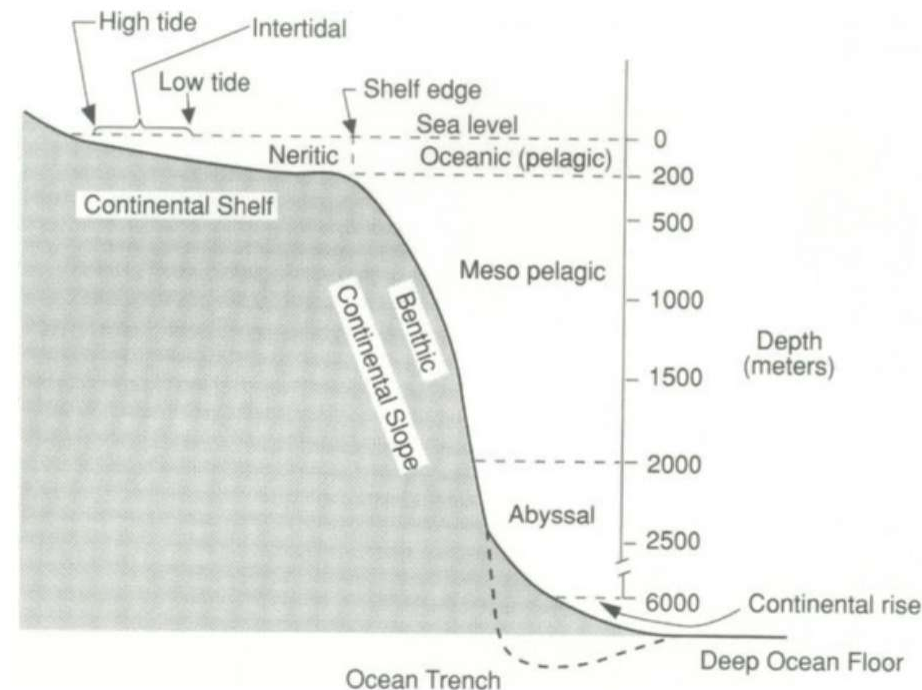
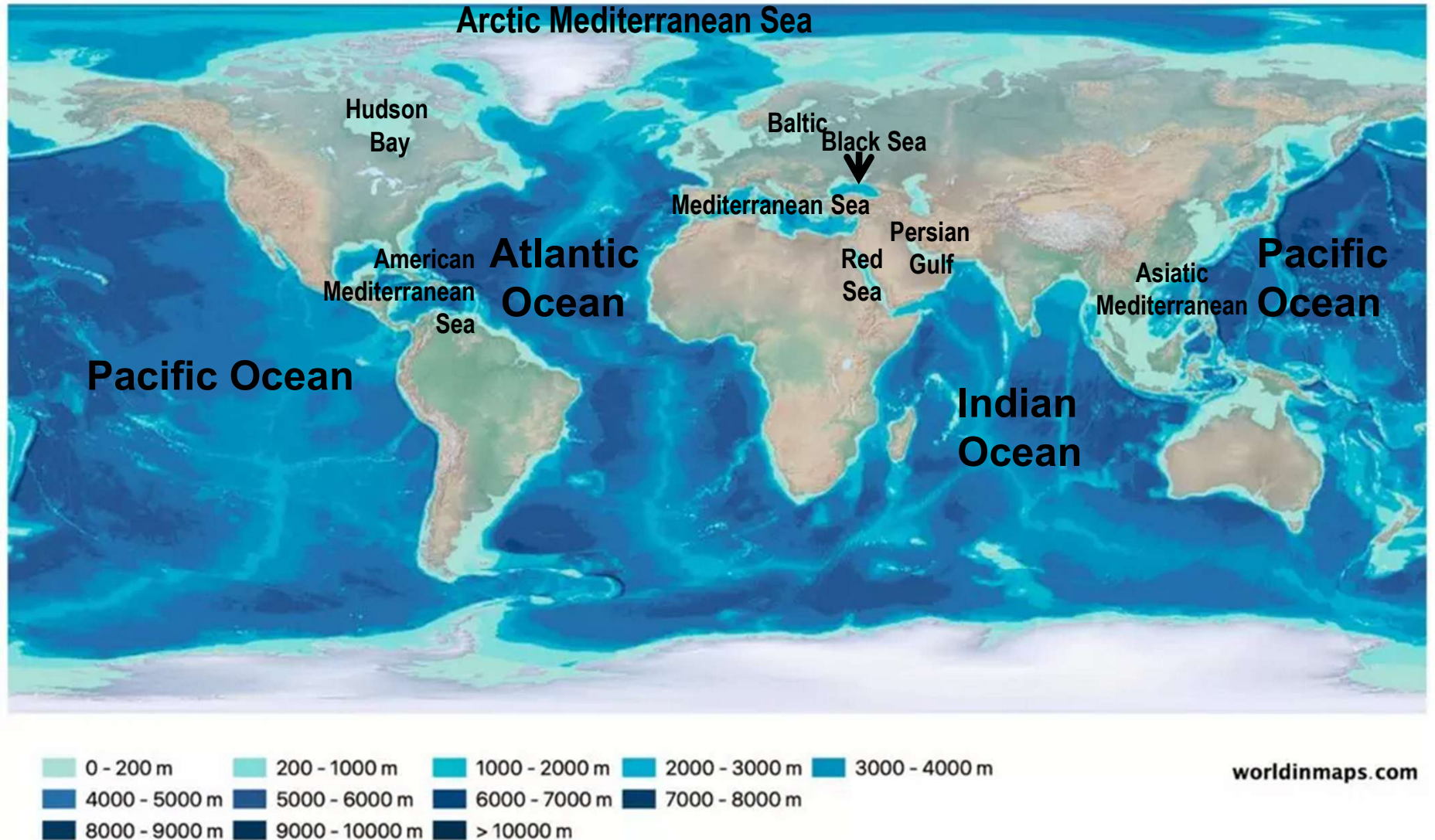


FIG. 10-1 Schematic representation of the physiographic profile at ocean margins.

Barymetric map



Mediterranean comes from Latin and means surrounded by land

Fact sheet major oceans

The Pacific is the largest ocean followed by the Atlantic and Indian Ocean.

The Pacific is also the deepest ocean as it has in relative terms more sea floor covered by old oceanic crust and less mid-ocean ridges. Marginal seas on shelves are shallow.

Table 10-1

Area, mean depth, and volume of oceans and seas

Region	Area (10 ⁶ km ²)	Mean depth (m)	Volume (10 ⁶ km ³)
Pacific Ocean	165.25	4282	707.56
Atlantic Ocean	82.44	3926	323.61
Indian Ocean	73.44	3963	291.03
Three oceans only	321.13	4117	1322.20
Arctic Mediterranean	14.09	1205	16.98
American Mediterranean	4.32	2216	9.57
Mediterranean Sea and Black Sea	2.97	1429	4.24
Asiatic Mediterranean	8.14	1212	9.87
Baltic Sea	0.42	55	0.02
Hudson Bay	1.23	128	0.16
Red Sea	0.44	491	0.21
Persian Gulf	0.24	25	0.01
Marginal seas	8.08	874	7.06
Three Oceans plus adjacent seas	361.06	3795	1370.32
Pacific Ocean	179.68	4028	723.70
Atlantic Ocean	106.46	3332	354.68
Indian Ocean	74.92	3897	291.94

Caribbean & Gulf of Mexico

South China Sea

Average topography

On average continents are 840 m above mean sea level, but the mode is ~ 400 m. There is a lot of land that is marginally above/below sea level which falls dry during glacial maxima and floods when the poles are ice free.

The average depth of the ocean is 3730 m but the mode is ~4200 m. There is very little land between 200 m and 3000 m below mean sea level and most of it is along mid-ocean ridges and the along the tracks of mantle hotspots. Some is located in marginal seas.

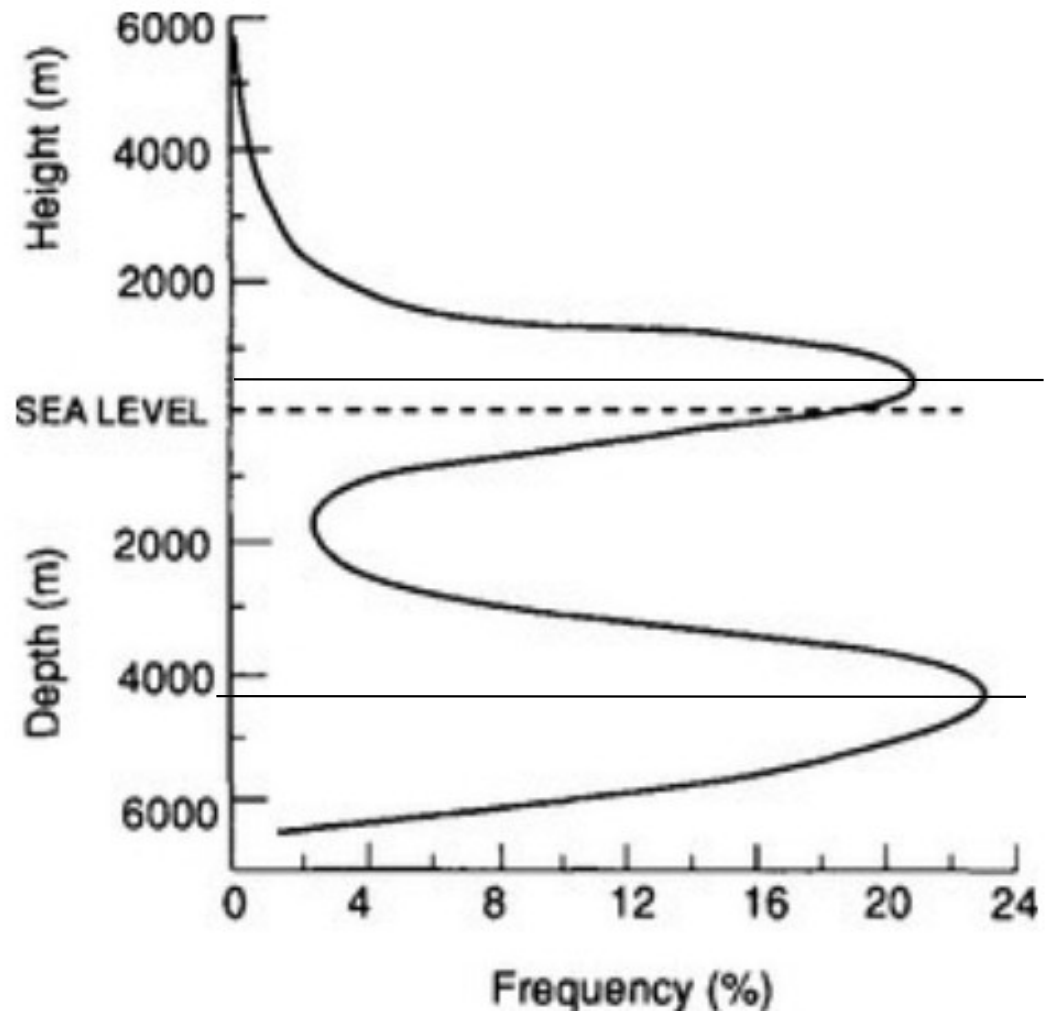


FIG. 10-2 A frequency distribution of elevation intervals of the Earth's surface.

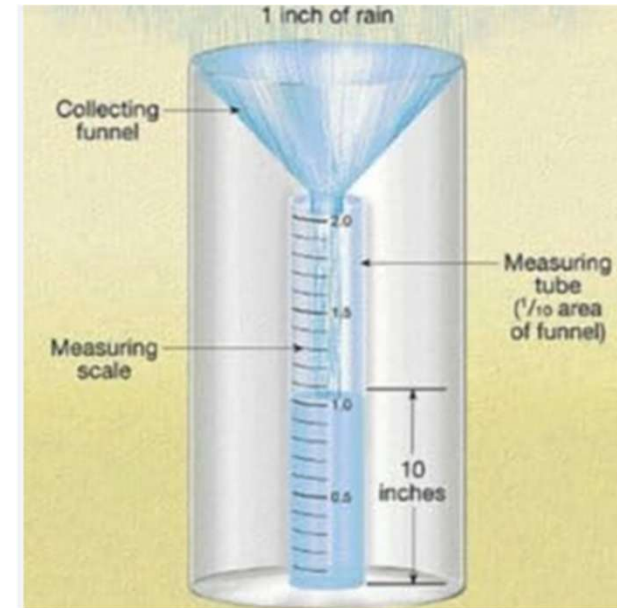
Fact sheet water balance of the major oceans

Table 10-2

A breakdown of the water balance for the four main ocean basins (cm/yr for the area of the respective basins)^a

Ocean	Precipitation	Runoff from adjoining land areas	Evaporation	Water exchange with other oceans
Atlantic	78	20	104	6
Arctic	24	23	12	35
Indian	101	7	138	30
Pacific	121	6	114	13

^aFrom M. I. Budyko (1958). "The Heat Balance of the Earth's Surface" (Trans. N. A. Stepanova). Office of Technical Services, Department of Commerce, Washington.



If the precipitation on the Atlantic is 78 cm yr⁻¹ then how much rain has fallen?

The area of the Atlantic is:
 $82.44 \times 10^6 \text{ km}^2$. So:
 $82.44 \times 10^6 [\text{km}^2] \times 78 \times 10^{-5} [\text{km yr}^{-1}]$
 $= 64303 [\text{km}^3 \text{ yr}^{-1}]$

Fact sheet chemical composition of the major oceans

Table 10-3

Average temperatures and salinity of the oceans, excluding adjacent seas^a

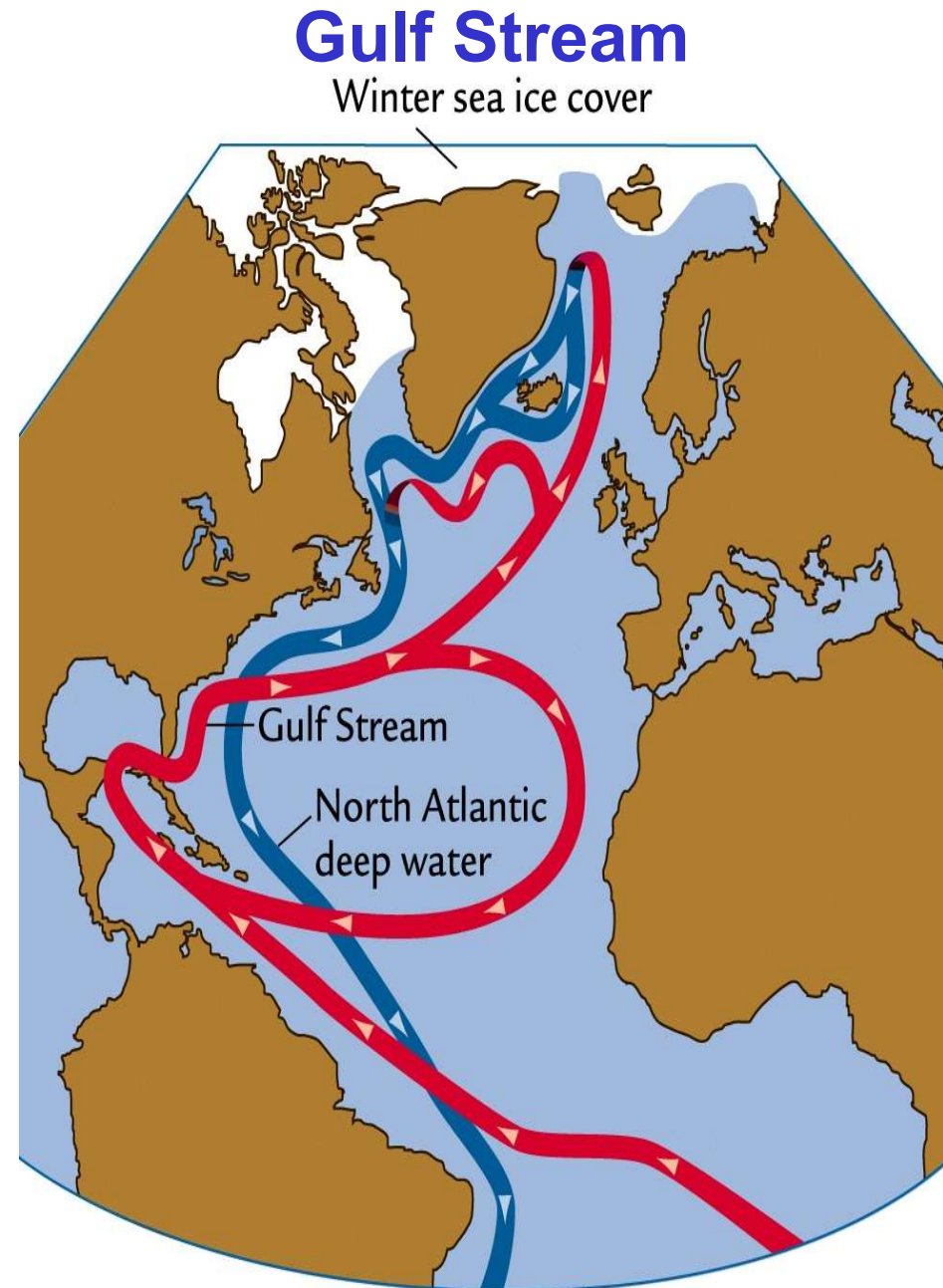
	Temperature (C)	Salinity (parts per thousand)
Pacific (total)	3.14	34.60
North Pacific	3.13	34.57
South Pacific	3.50	34.63
Indian (total)	3.88	34.78
Atlantic (total)	3.99	34.92
North Atlantic	5.08	35.09S
South Atlantic	3.81	34.84
Southern Ocean ^b	0.71	34.65
World ocean (total)	3.51	34.72

Salinity is expressed in gram of salt per kg of sea water or permille ‰

Warmer oceans in arid zones tend to be more saline because more water evaporates from them, but freshwater input (rainfall, rivers, ice melt) also matter

Ocean circulation

- Pole ward flowing warm surface currents like the Gulf stream and the Kuroshiro current have profound effect on the sea surface temperature (SST) and the climate of land areas bordering the oceans.
- The Gulf stream transports approximately 3.2 PW of heat to the North Atlantic moderating the climate of northern Europe



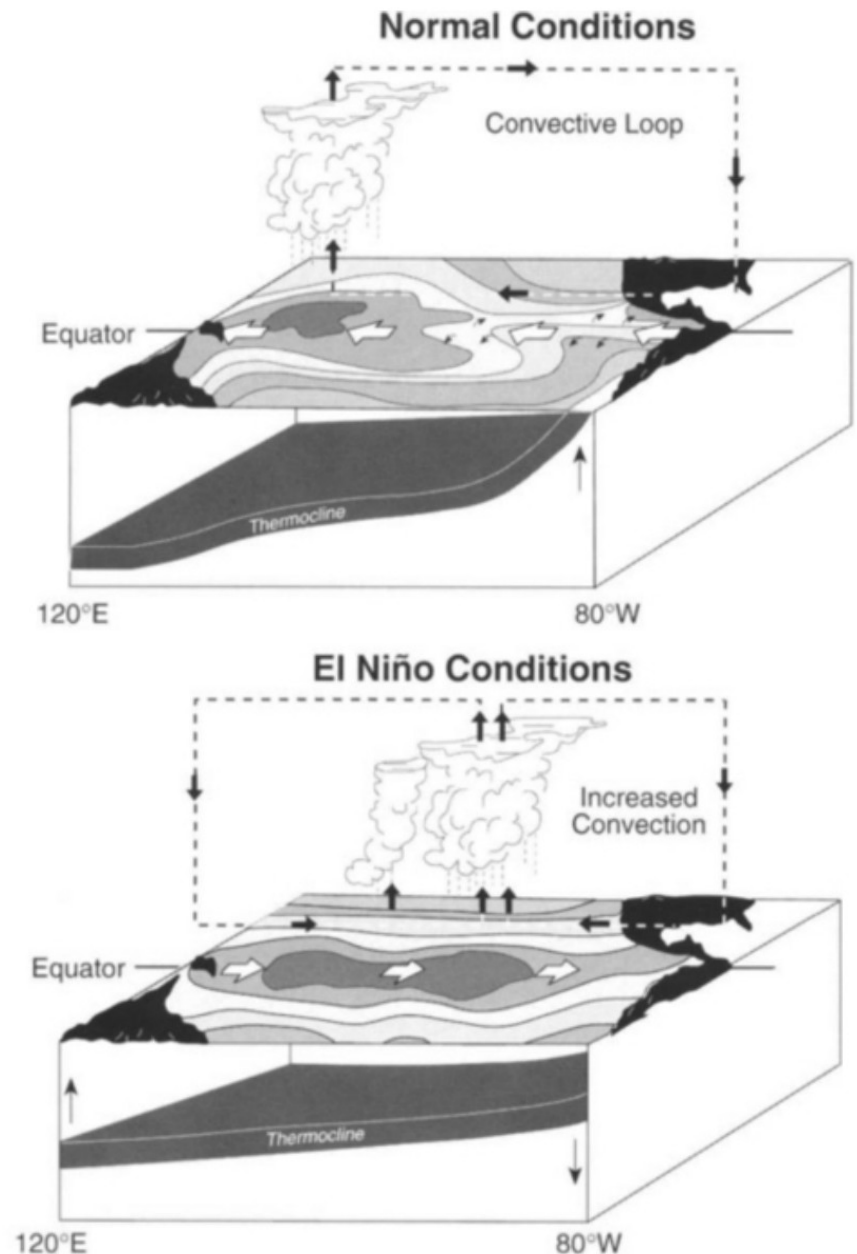
Ocean circulation

- **El-Niño southern oscillation (ENSO) phenomenon is an interannual perturbation of the climate system characterized by periodic weakening of the trade winds and warming of the surface water in the central and eastern equatorial Pacific Ocean**
- **This disrupts atmospheric circulation and weather patterns and leads to excessive rain along the coast of South America and dry conditions in Indonesia.**

FIG. 10-7 Schematic of normal and El Niño conditions in the equatorial Pacific.

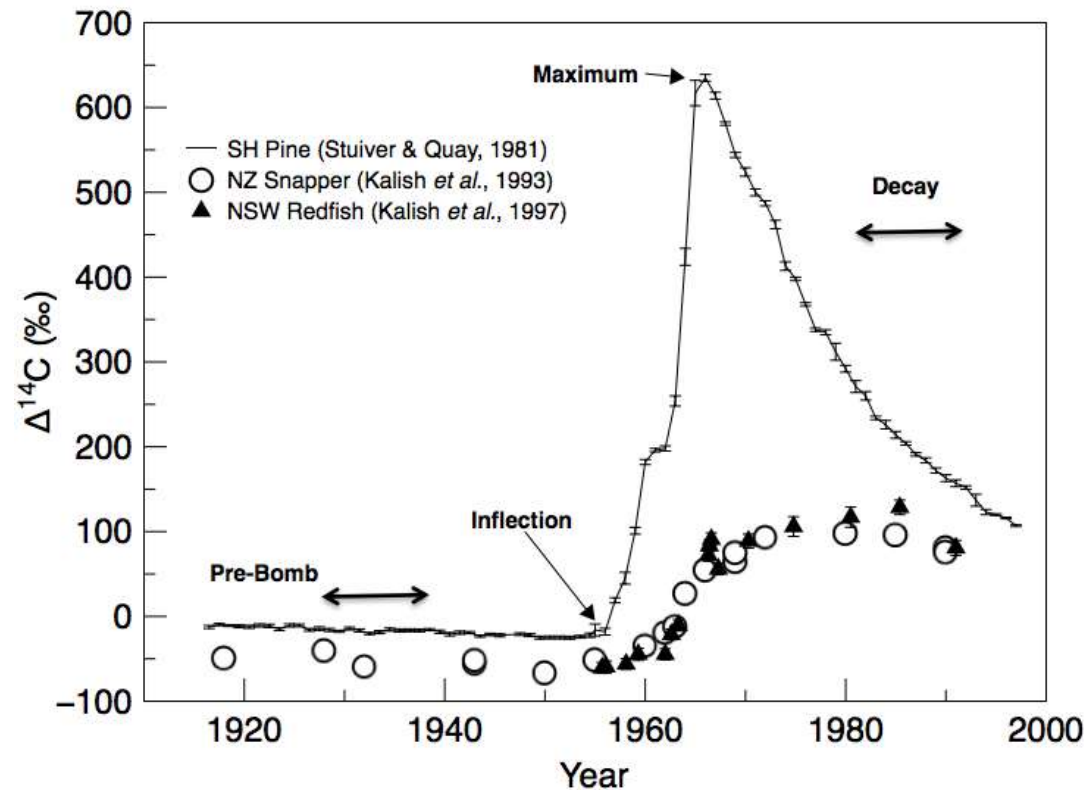
(Figure kindly provided by Dr Michael McPhaden of NOAA.)

El-Niño southern oscillation



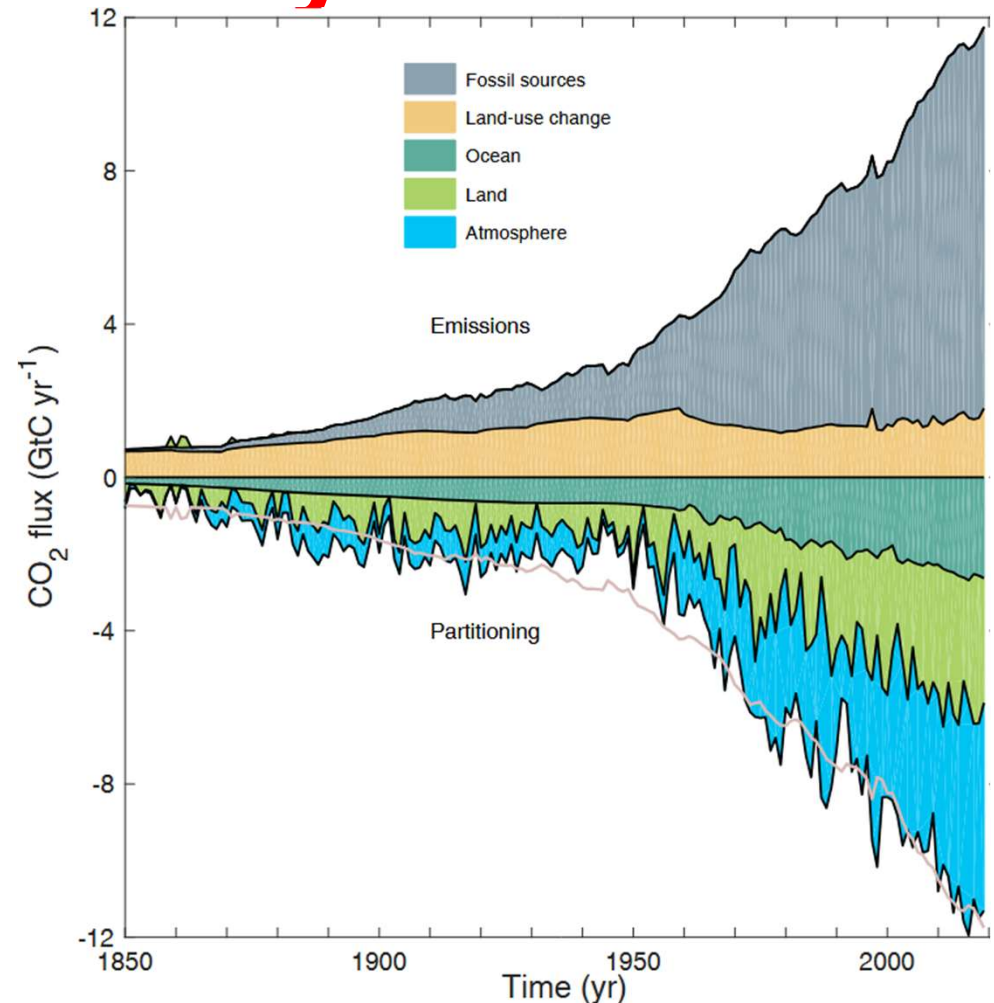
The golden Peak is used to study ocean circulation system

- Atmospheric testing of nuclear bombs resulted in contamination of the atmosphere and surface ocean with various radioactive isotopes including ^{14}C ($t_{1/2}=5730$ yr), ^3H ($t_{1/2}=12.43$ yr), ^{90}Sr ($t_{1/2}=29.1$ yr), ^{239}Pu ($t_{1/2}=24,100$ yr) and ^{240}Pu ($t_{1/2}=6,560$ yr).
- These isotopes are slowly being mixed through the ocean and can be used as radioactive dyes and clocks.



The golden Peak is used to study ocean circulation system

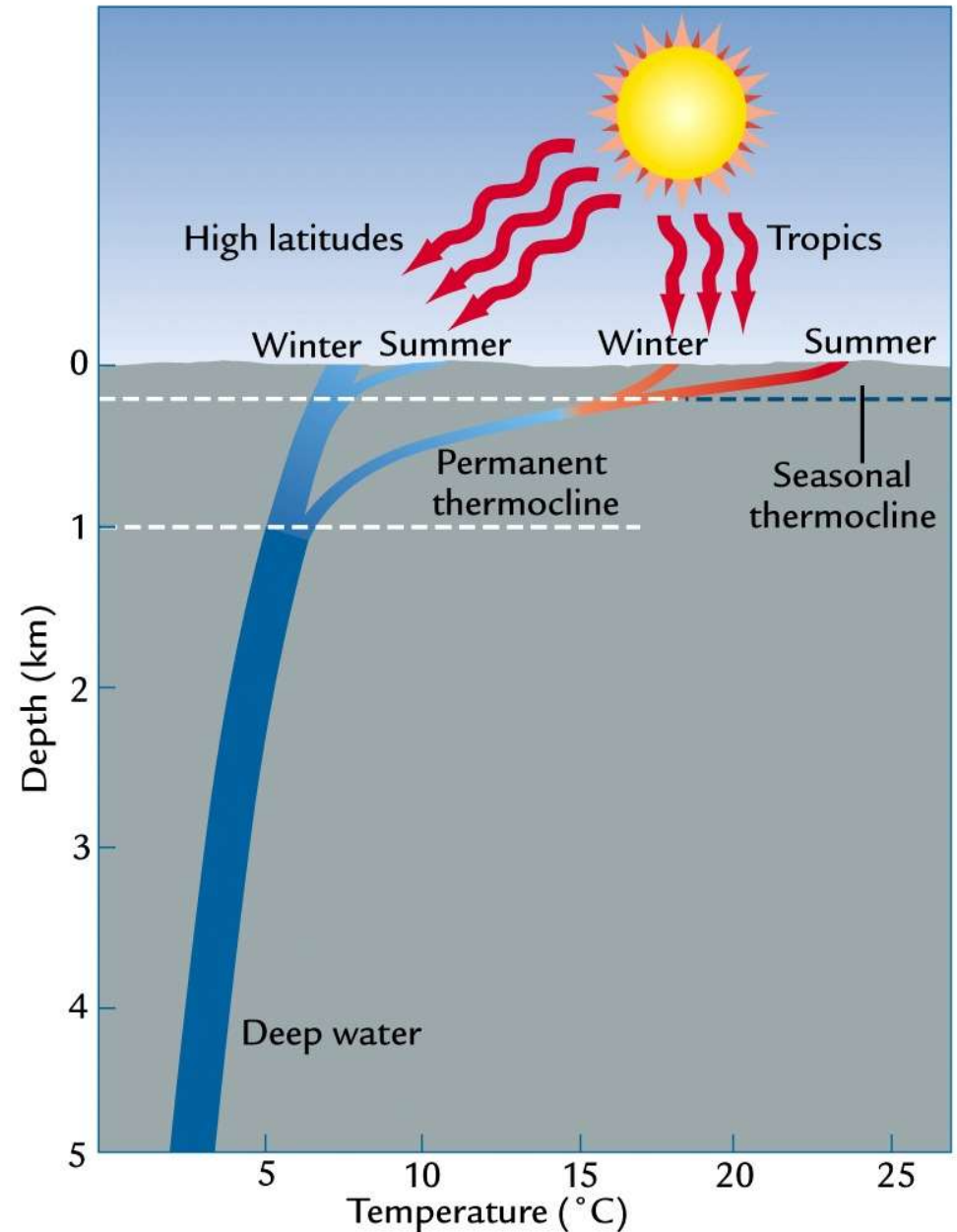
- The atmospheric CO₂ concentration has increased at a rate of $\sim 3.2 \text{ Gt C yr}^{-1}$ over the past decades. That is much less than the increase of CO₂ emissions from anthropogenic emissions and deforestation ($\sim 7.0 \text{ Gt C yr}^{-1}$). Some of the CO₂ has been taken up by the ocean.
- CO₂ taken up by the ocean is transported into the intermediate and deep water by the subduction of water masses. Circulation replenishes the surface with water under saturated with respect to the anthropogenically perturbed CO₂ levels



Friedlingstein, et al. Global Carbon Budget 2020, Earth Syst. Sci. Data, 12, 3269–3340, <https://doi.org/10.5194/essd-12-3269-2020>, 2020.

Density stratification in the ocean

- The density of water is controlled by temperature and salinity.
- The surface layer is the region from the sea surface to the depth having a temperature of about 10°C .
- The transition region where temperature decreases from 10°C to 4°C is called the thermocline. Water density is highest at 4°C .
- The deep sea is the region below the thermocline.
- Since the warm water at surface is less dense than the colder water below the water column is inherently stable with slow vertical mixing
- Water has a high heat capacity. Lots of heat required for a small change in temperature.
- Lateral temperature and salinity differences are small over large areas due to currents.



Density stratification and ventilation in the ocean

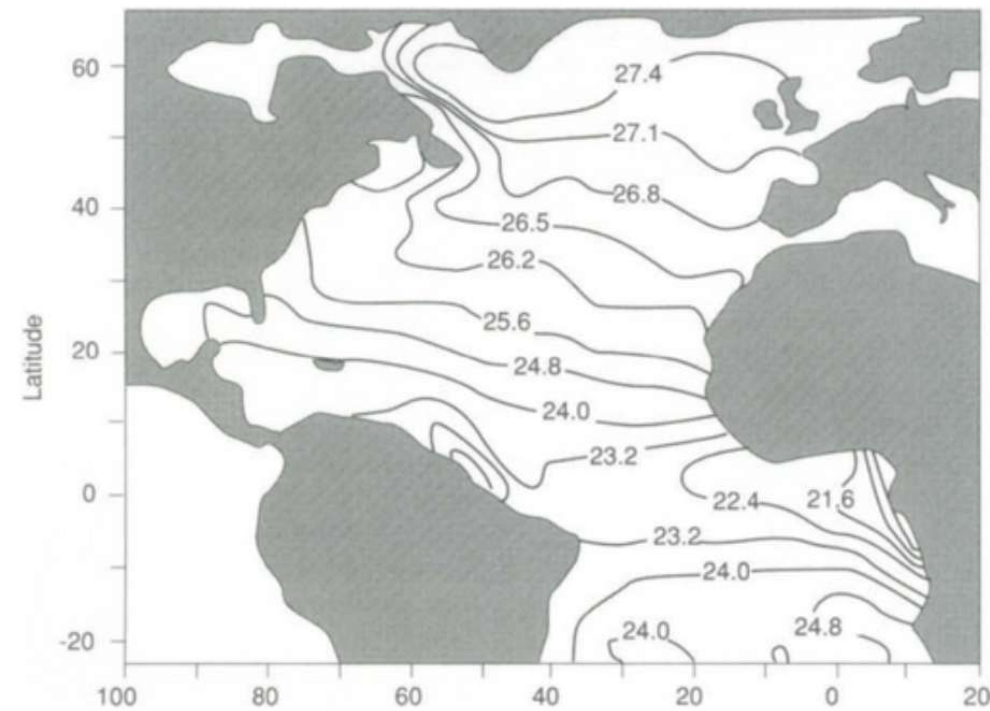


FIG. 10-9 Map of winter outcrops of isopycnal surfaces in the Atlantic Ocean.

- The advection as well as ventilation and mixing of the ocean takes place along isopycnal (equal density) surfaces.
- Continuous upwelling happens near the equator on the eastern side of the large oceans (west coast of large continents) due to diverging trade winds driving the surface water against east coast where it diverges north and south. Downwelling happens mostly near the north pole in the North Atlantic.
- The ventilation is very important for understanding the fate of excess CO₂ in the atmosphere.

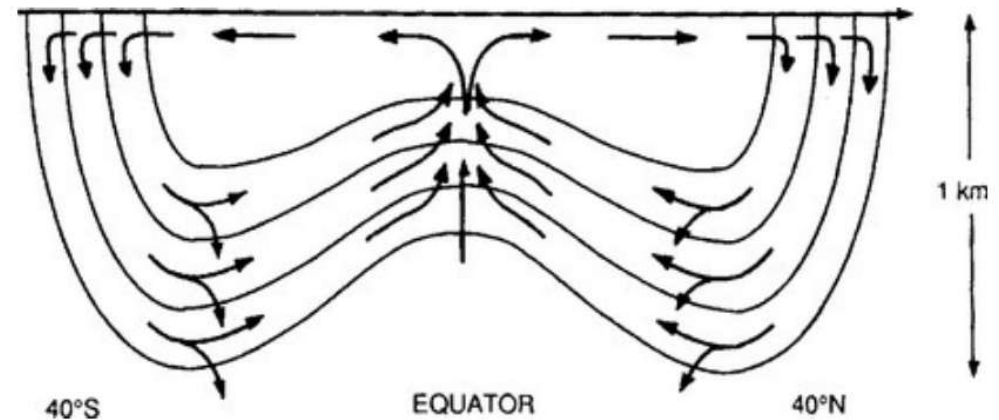
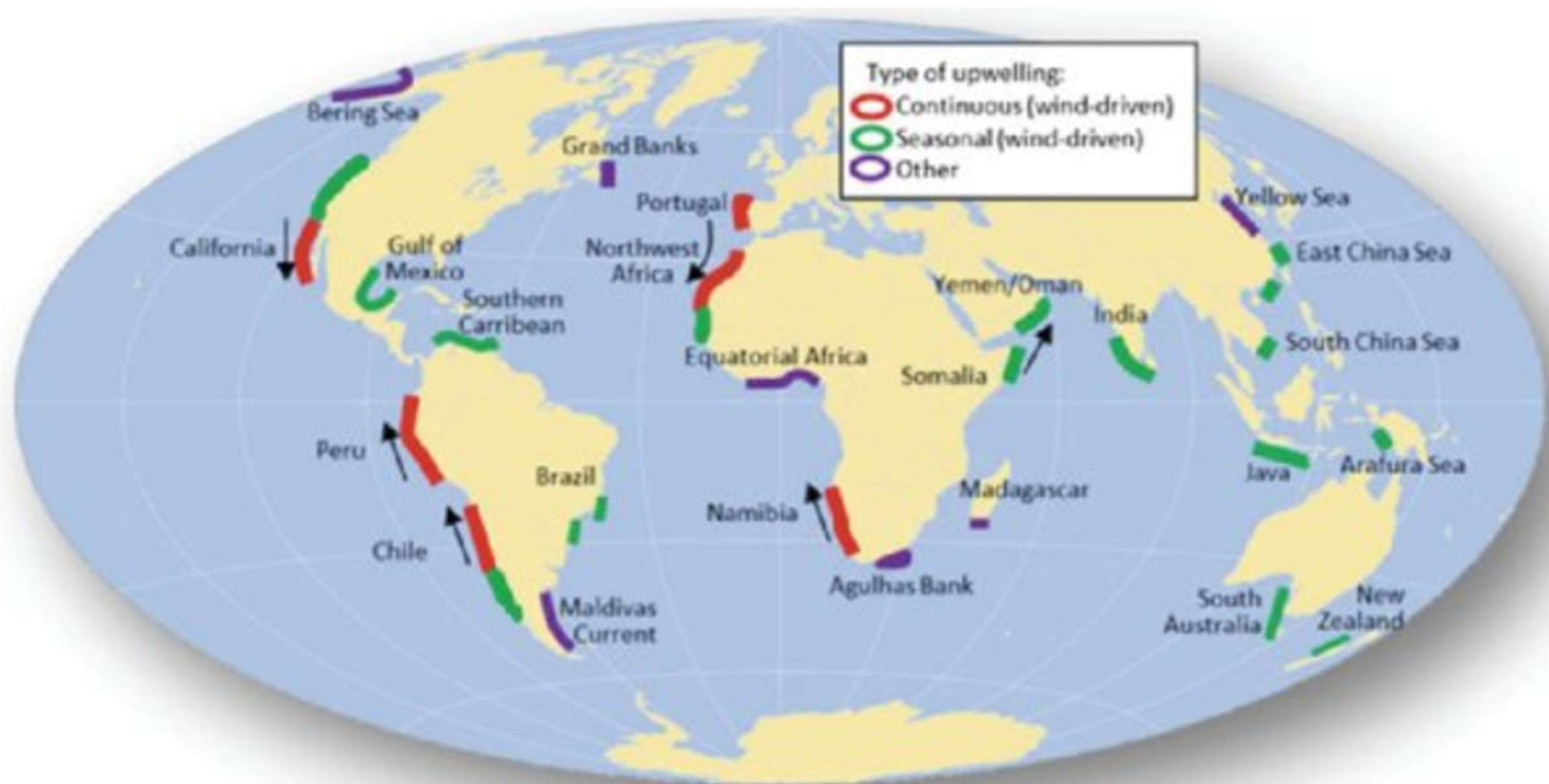
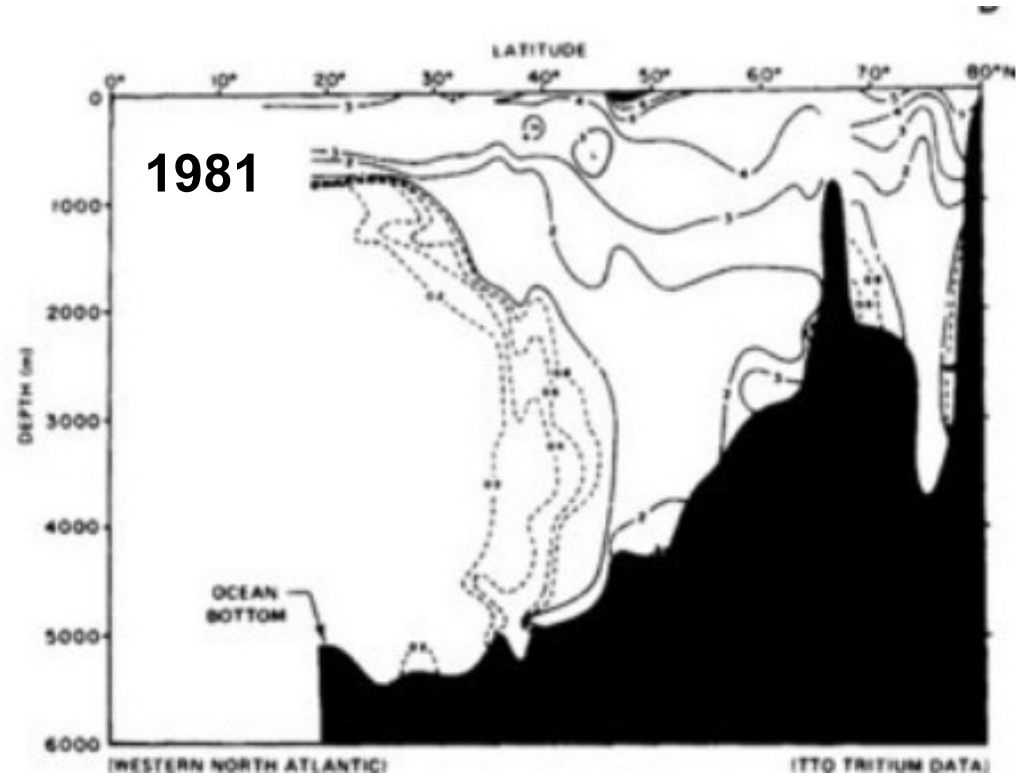
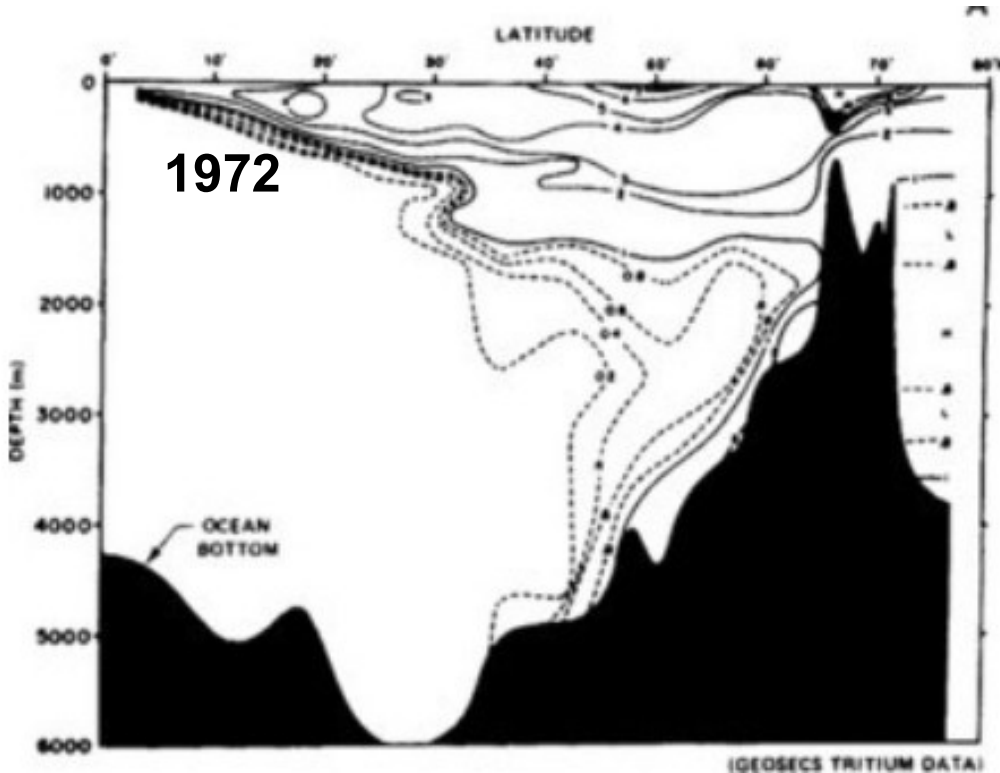


FIG. 10-8 The pathways followed by the water ventilating the main oceanic thermocline. (Reproduced with permission from W. S. [Broecker and T.-H. Peng \(1982\)](#) . "Tracers in the Sea," p. 440, Eldigio Press, Palisades, NY.)

Upwelling of bottom water



Ventilation studies with the Nuclear testing peak



- Dept profiles from cruises in 1972 (GEOSECS) and 1981 (TTO) show the intrusion of ^3H into the thermocline and the rapid ventilation of the North Atlantic.
- The bomb tests peaked in 1960.

FIG. 10-10 Tritium section of the western Atlantic from 80°N to the equator versus depth (m). Vertical exaggeration is 2000:1. Horizontal scale is proportional to cruise track. (Reproduced with permission from H. G. [Ostland and R. A. Fine \(1979\)](#) . Oceanic distribution and transport of tritium. In "Behaviour of Tritium in the Environment" (Proceedings of a Symposium, San Francisco, 16–20 October 1978,

Classified by volume, most water in the ocean occurs in a narrow range of T and Salinity

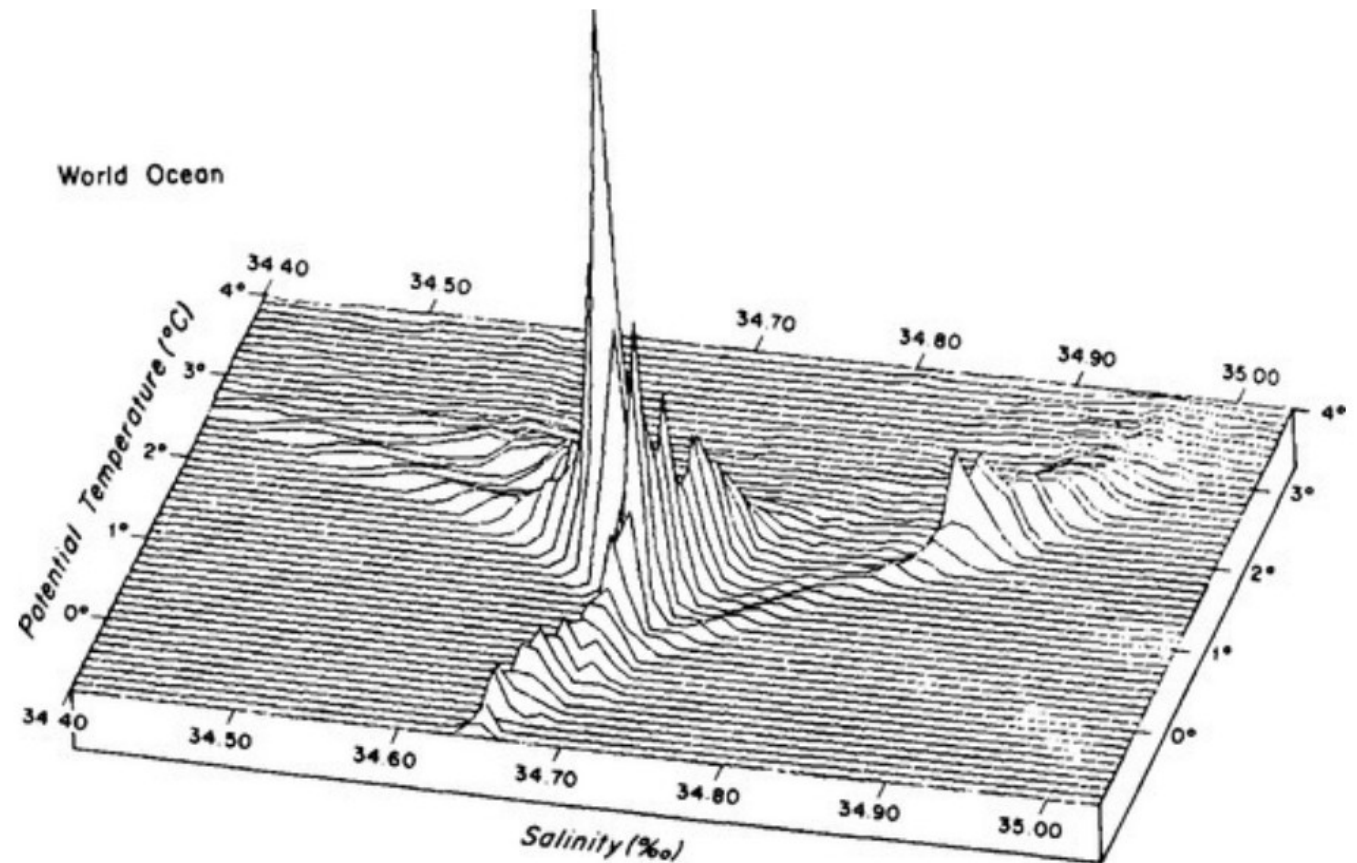


FIG. 10-3 Simulated three-dimensional T-S diagram of the water masses of the world ocean. Apparent elevation is proportional to volume. Elevation of highest peak corresponds to $26.0 \times 10^6 \text{ km}^3$ per bivariate class $0.1^\circ\text{C} \times 0.01\text{‰}$. (Reproduced with permission from L. V. Worthington, The water masses of the world ocean: some results of a fine-scale census. In B. A. Warren and C. Wunsch (1981). "Evaluation of Physical Oceanography," MIT Press, Cambridge, MA.)

Temperature & Salinity diagrams for major currents in the main

- Most currents have unique T & S signatures. The density characteristics are locked into a parcel of water when it is at the sea surface. They depend on the balance between evaporation and precipitation for salinity & the heat received from solar input for temperature.
- The signature is preserved when water sinks and modified by mixing with water from other sources.

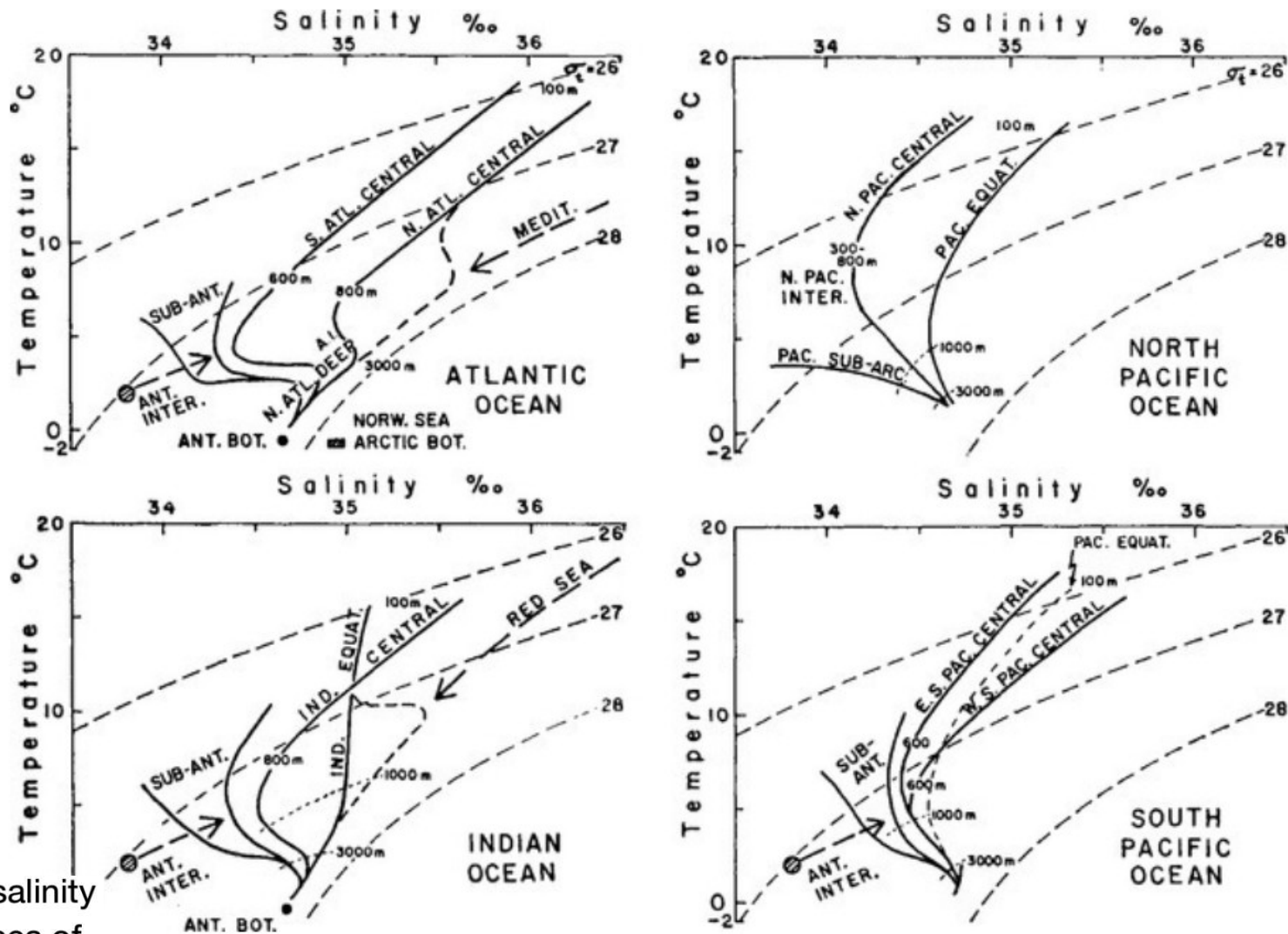
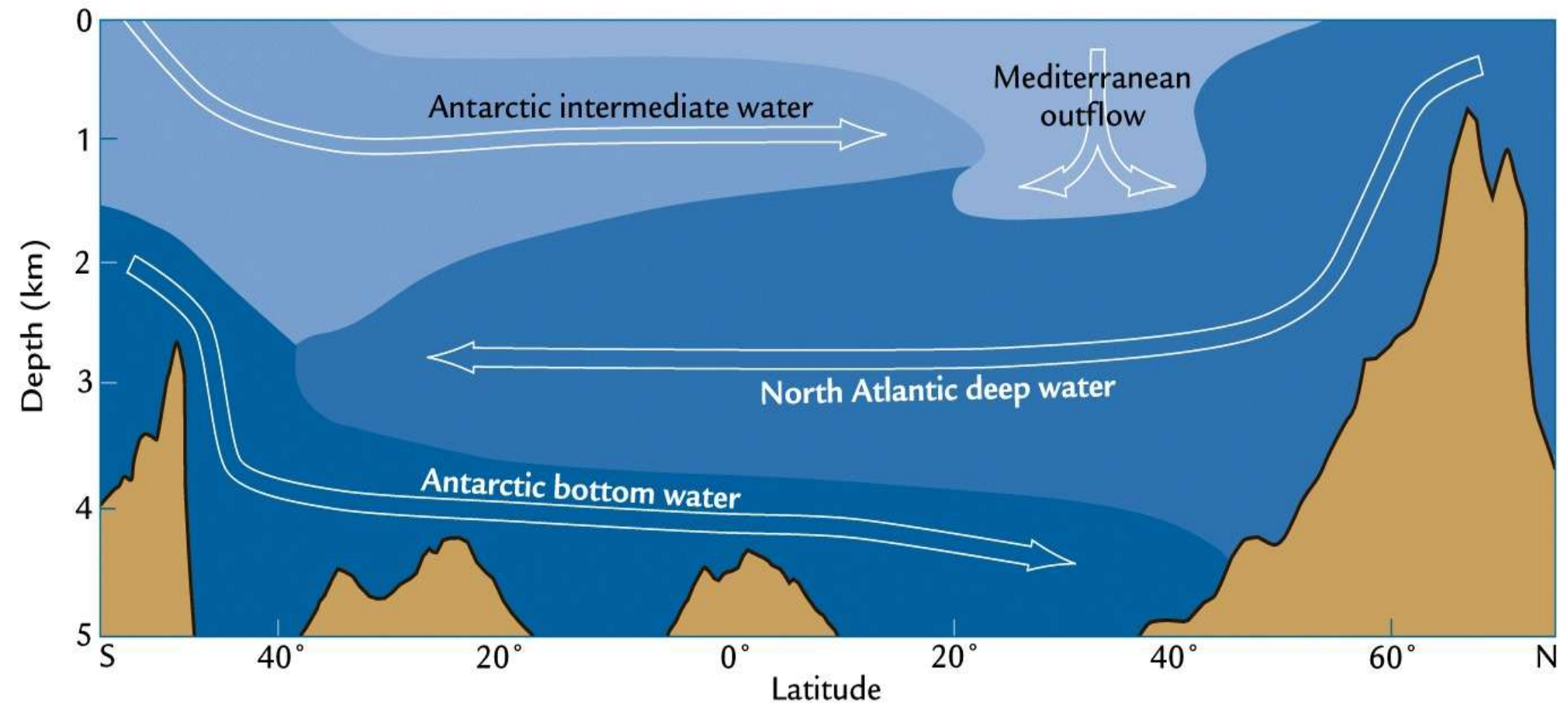


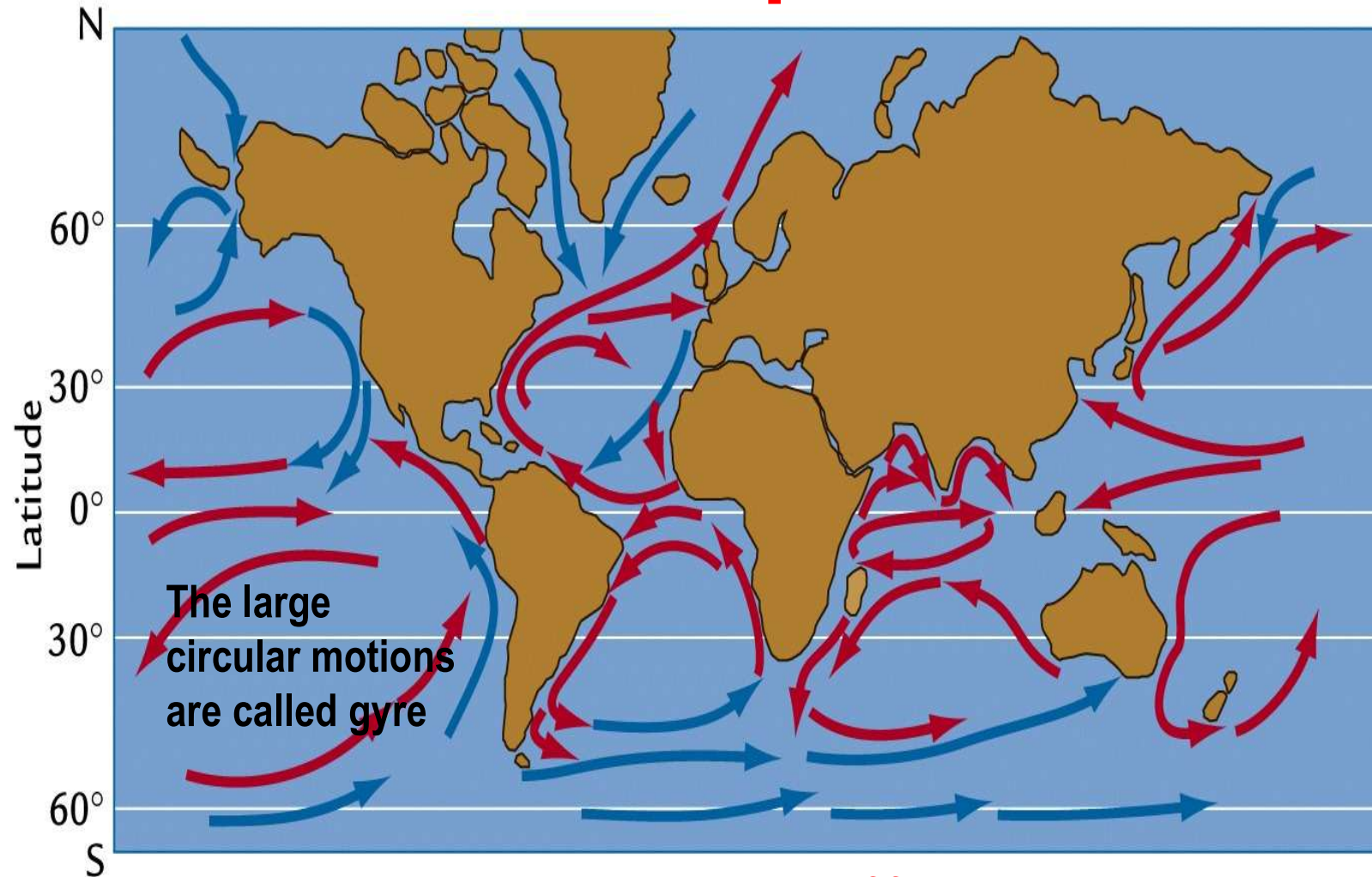
FIG. 10-4 Average temperature/salinity diagrams for the main water masses of the Atlantic, Indian, and Pacific Oceans.

(Reproduced with permission from G. L. [Pickard and W. J. Emery \(1982\)](#) . "Descriptive Physical Oceanography," pp. 138-139, Pergamon Press.)

Vertical structure of major currents example Atlantic



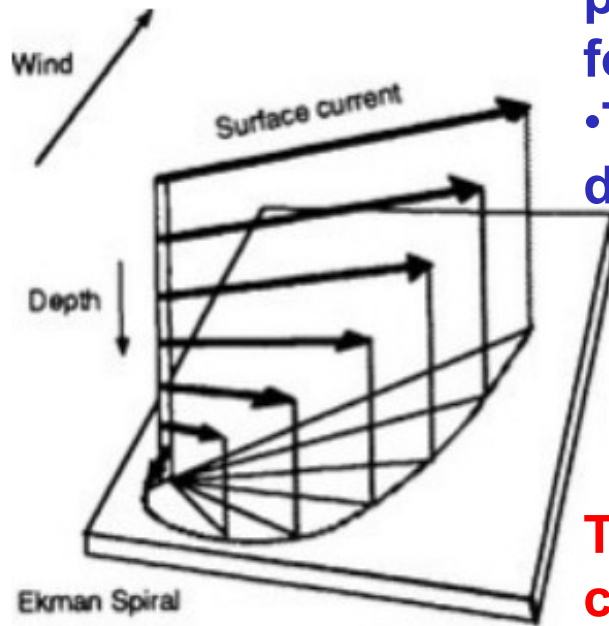
Horizontal transport in the ocean



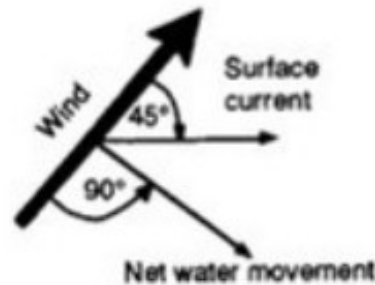
- Surface ocean currents are driven by winds. As a result of friction, winds drag the ocean surface and induces water movement in the upper ~100 m in the direction of the wind. In well-developed currents (Examples, Gulf Stream, Kuroshiro Current) this motion may persist to a depth of ~1-2 km
- Coriolis effect deflects water to right in N. hemisphere and to the left in S. hemisphere.
- Trade winds move water against the eastern coast of continents where it is deflected North and South. It cools and sinks in the Arctic & Antarctic. Western continental margins have upwelling waters.

Eckman Spiral

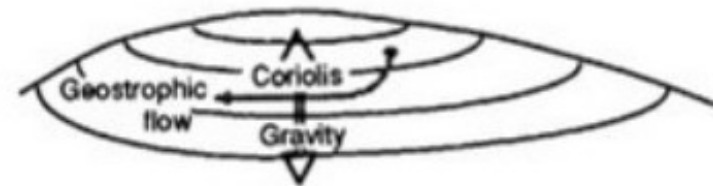
- In well developed gyres like the North Pacific water is pushed to the center of the gyre (25°N) and piles up to form a sea surface topographic high.
- The water then flows downhill due to gravity, but is deflected due to the Coriolis force.



(a)



(b)



(c)

The net current is clockwise in the Northern; and counter-clockwise in the Southern Hemisphere and parallel to the contours.

- The wind drags the surface and sets it in motion and this layer drags the following layer.
- As a result of friction between layers each layer moves more slowly than the one above and its motion is deflected clockwise in the northern hemisphere by the Coriolis force. This feature is called the **Eckman Spiral**.

FIG. 10-5 Sketch of (a) current vectors with depth characteristic of an Eckman spiral; (b) relationship between wind, surface current, and net water movement vectors; and (c) production of circular gyres from the net interaction of the Coriolis force and Eckman transport.

Horizontal transport in the surface ocean

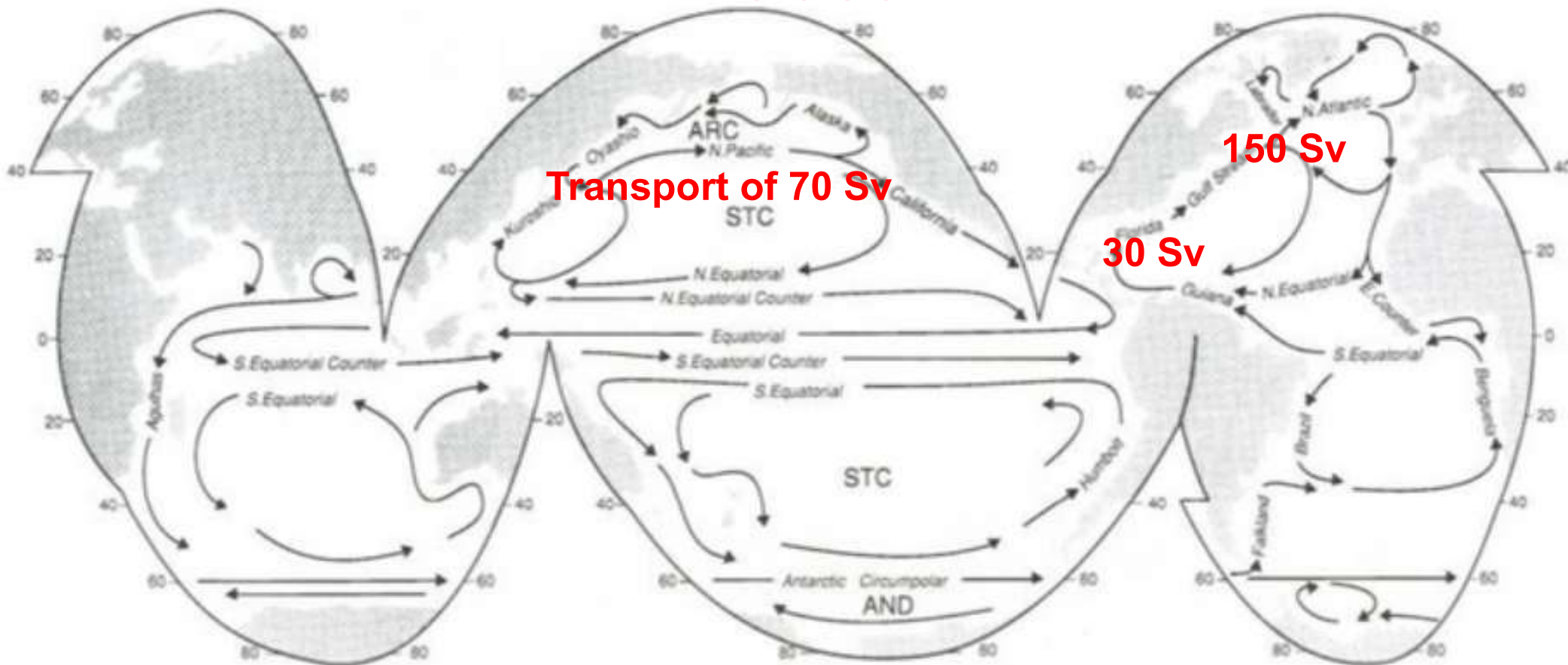


FIG. 10-6 Global map of major ocean currents. AND = Antarctic Divergence; STC = subtropical convergence; ARC = Arctic convergence.

1 Sv (Sverdrup)
= $1 \times 10^6 \text{ m}^3/\text{s}$

The ocean conveyor belt

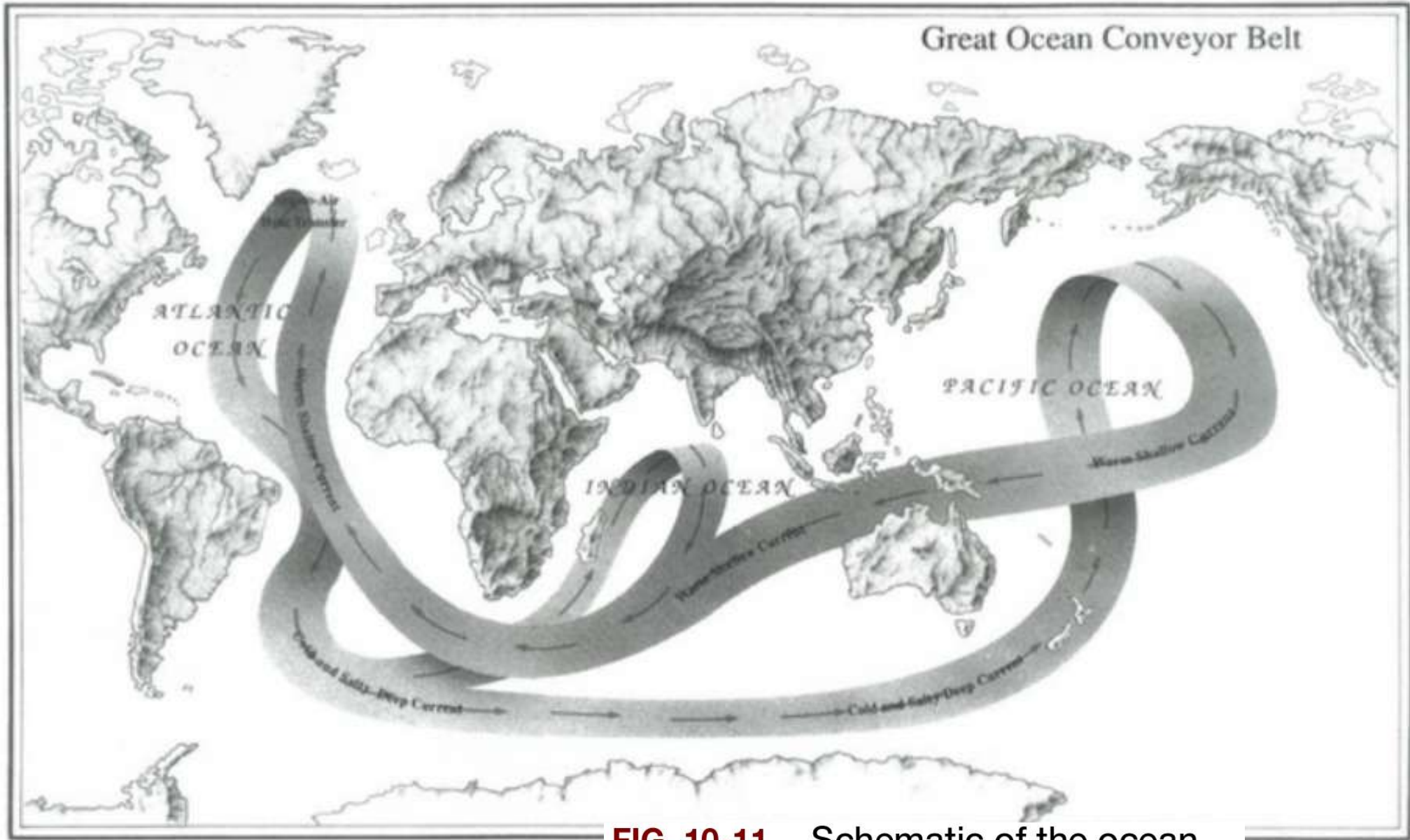
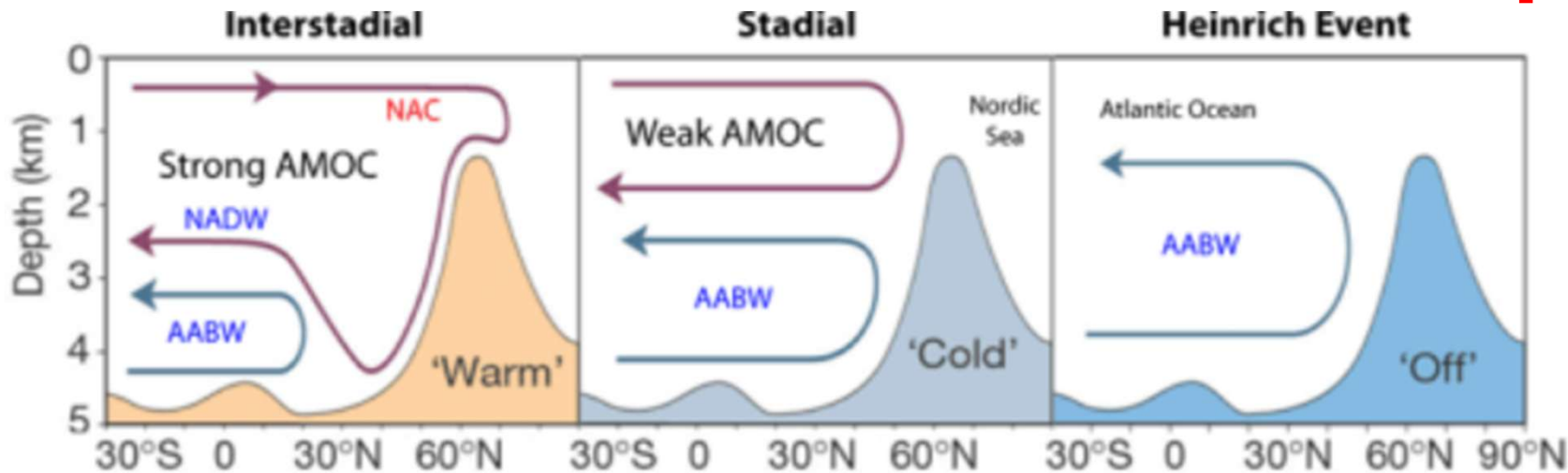


FIG. 10-11 Schematic of the ocean “conveyor belt” from [Broecker \(1991\)](#).

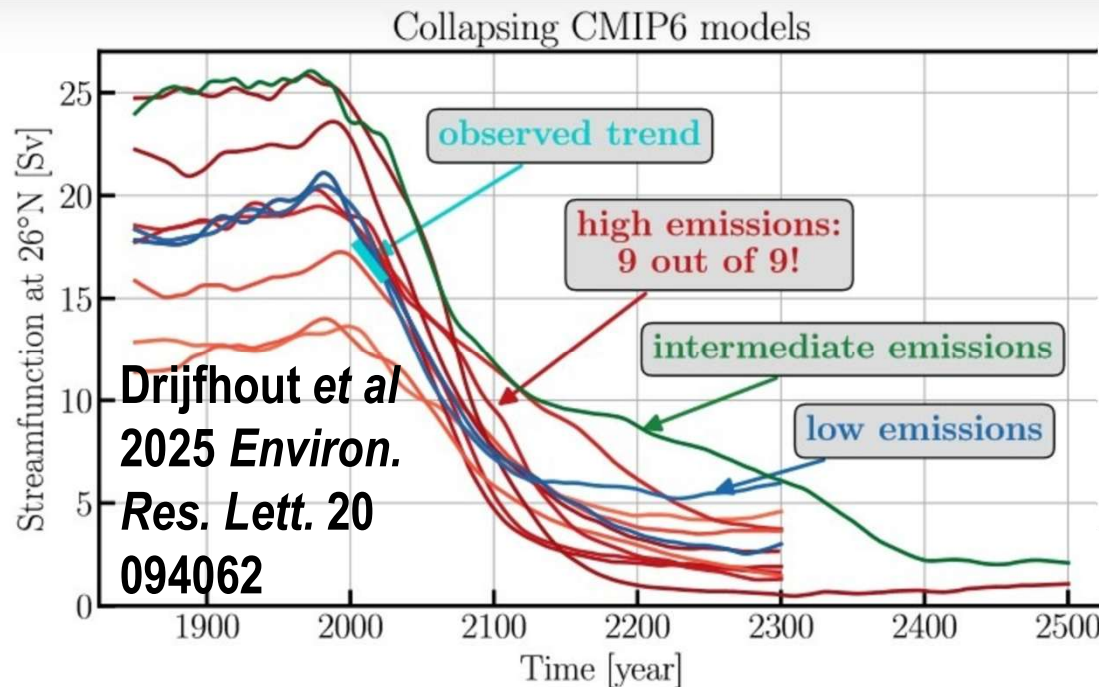
(Reproduced with permission of the illustrator, Joe Le Monnier.)

The ocean conveyor belt is sensitive to freshwater inputs

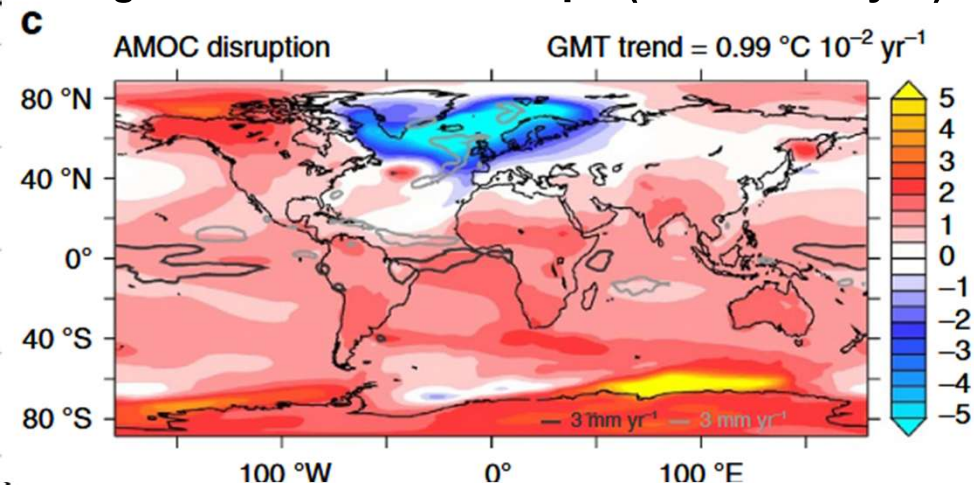


There have been multiple disruptions in the Pleistocene

Switching off the Gulf stream drastically changes climate over Europe (colder & dryer)



Drijfhout et al
2025 Environ.
Res. Lett. 20
094062



Sgubin et al. 2018 Nature
Communications 8:14375

PRACTICE QUESTIONS

You can submit the answer to these questions in the next tutorial

MCQs

1. The densest (heaviest) ocean water has:
 - a. warm temperature, high salinity
 - b. warm temperature, low salinity
 - c. cold temperature, low salinity
 - d. cold temperature, high salinity
2. During El niño events, the _____ equatorial Pacific Ocean are unusually warm.
 - a. western
 - b. eastern and central
 - c. western and central
 - d. entire
3. Upwelling and cold surface ocean currents occur generally along:
 - a. mid-latitude west coasts of continents
 - b. mid-latitude east coasts of continents
 - c. both mid-latitude coasts of continents
 - d. none of these
4. Surface ocean currents are driven by:
 - a. the Earth's rotation
 - b. density differences
 - c. winds
 - d. tides

MCQs

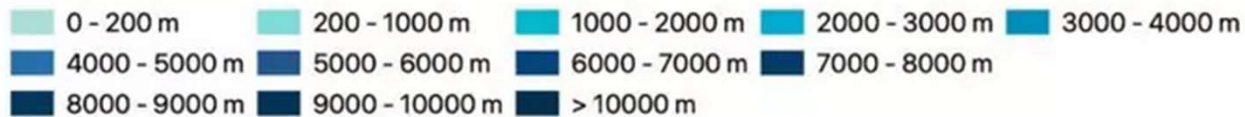
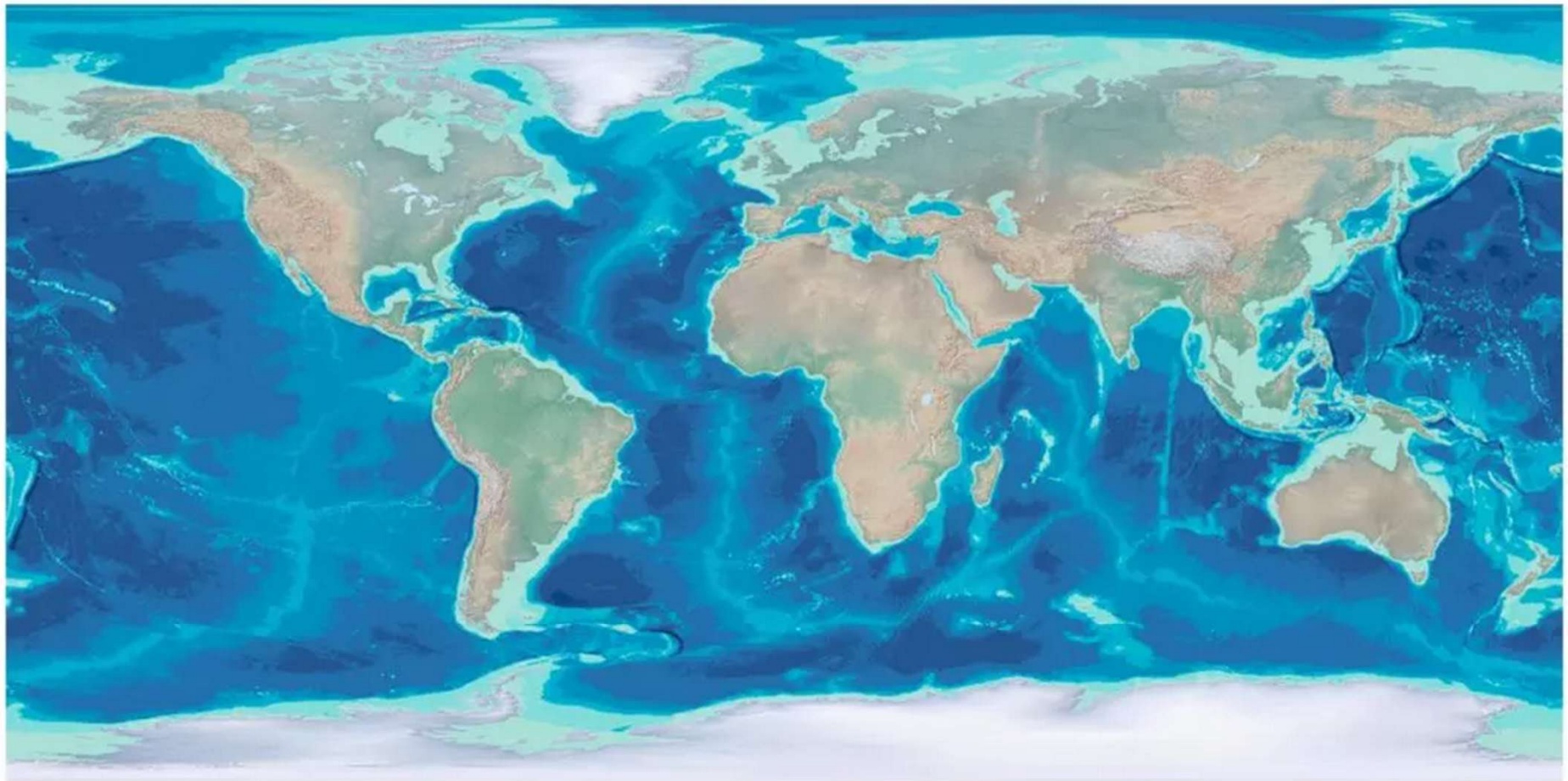
1. The thermocline is a region:
 - a. of warm surface water
 - b. cold deep water
 - c. of rapid change from warm to cold water
 - d. rapid salinity changes
2. In normal years, the _____ region is the warmest portion of the equatorial Pacific Ocean.
 - a. western
 - b. eastern
 - c. central
 - d. entire
3. Deep ocean circulation, that often resembles a conveyor belt, is called:
 - a. the thermocline
 - b. the thermohaline circulation
 - c. the circulation of the 7 seas
 - d. pycnocline circulation
4. What is the net movement of water in an infinitely deep spiral Ekman transport system?
 - a. At a 90-degree angle to the wind direction
 - b. In the same direction as the wind
 - c. In the opposite direction of the wind
 - d. In a random direction

Numerical

1. If the precipitation on the Pacific is 121 cm yr^{-1} then how much rain has fallen? Its area is $179.66 \times 10^6 \text{ km}^2$
2. If the Pacific receives 6 cm runoff from rivers then how much water does it receive? Its area is $179.66 \times 10^6 \text{ km}^2$
3. If ocean water contains 35,000 ppm dissolved salts then what is its salinity?

Long answer style

1. Label the major oceans of the world in the following map



worldinmaps.com

Long answer style

1. Sketch how the El Nino Southern oscillation perturbs Pacific sea surface temperatures, thermocline and rainfall patterns
2. Explain why the North Atlantic Overturning Circulation is very important and can drastically alter temperatures over Europe over relatively short (decadal) timescales.
3. What is the golden peak and how has it been used to study ocean circulation and deep water formation?
4. Why is the oceanic water column inherently stable.